Business Problem Statement

Modern retail markets move fast, and businesses often struggle to make sense of the massive volumes of transaction logs they collect every day. Without proper insights, management cannot easily figure out which customer segments drive the core revenue or how external variables like seasonal shifts, shipping preferences, and promotional discounts actually shape consumer buying habits.

To tackle this, our objective was to take raw retail logs and turn them into clear, strategic intelligence:

- The Dataset: Comprises 3,900 customer transactions spread across 20 distinct attributes.

Question: "How can a retail organization harness end-to-end data analytics to decode purchasing behaviors, optimize product inventory, and unlock targeted growth opportunities?"

Project Deliverables:

To solve this problem, we followed a structured 4-step data analytics workflow:

1. Data Cleaning & Preparation (Python):

- Imported the dataset using Pandas
- Checked for missing values (like filling missing review ratings with category medians) and created new columns like `age_group` for better grouping

2. Database Analysis & Querying (SQL):

- Loaded the cleaned data into a database (`customer shopping.sql`)
- Wrote SQL queries to find key business answers, such as revenue generated by gender, top-rated products, repeat buyers, and customer segmentation.

3. Interactive Dashboard (Power BI):

- Built a visual dashboard (`Customer Shopping Dashboards.pbix`) with interactive charts, season trends, category breakdowns, and subscription metrics for stakeholders.

4. Business Insights & Reporting:

- Summarized all findings, statistics, and strategic recommendations into a professional PDF report (`Customer Shopping Analysis Report.pdf`).